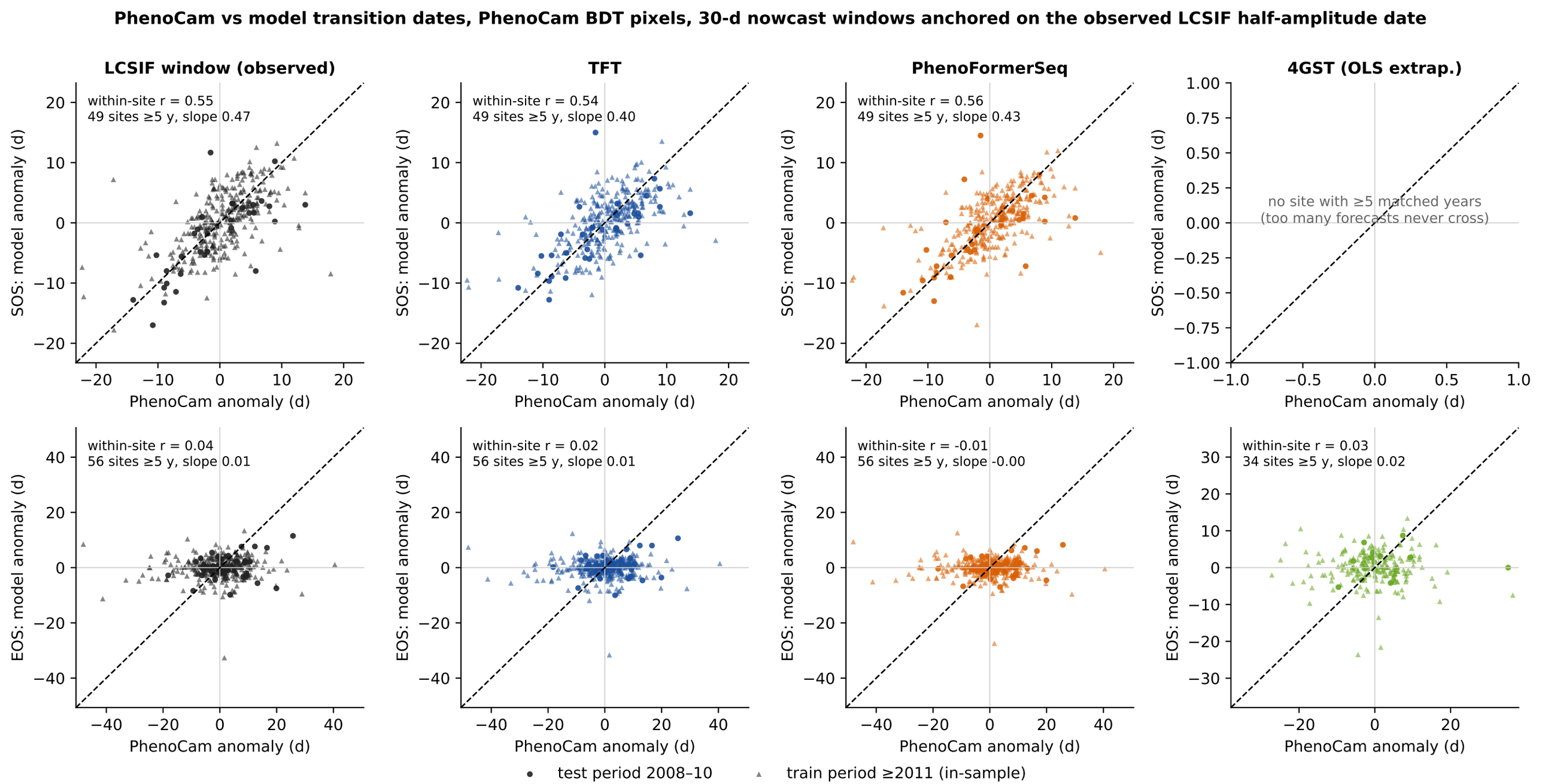

PhenoCam vs model transition dates, PhenoCam BDT pixels, 30-d nowcast windows anchored on the observed LCSIF half-amplitude date
LCSIF window (observed)
TFT
PhenoFormerSeq
4GST (OLS extrap.)
within-site r = 0.55
49 sites ≥5 y, slope 0.47
within-site r = 0.54
49 sites ≥5 y, slope 0.40
within-site r = 0.56
49 sites ≥5 y, slope 0.43
no site with ≥5 matched years
(too many forecasts never cross)
within-site r = 0.04
56 sites ≥5 y, slope 0.01
within-site r = 0.02
56 sites ≥5 y, slope 0.01
within-site r = -0.01
56 sites ≥5 y, slope -0.00
within-site r = 0.03
34 sites ≥5 y, slope 0.02
SOS: model anomaly (d)
EOS: model anomaly (d)
PhenoCam anomaly (d)
test period 2008–10
train period ≥2011 (in-sample)
2011–2021 lie in the training years of the TFT and PhenoFormerSeq checkpoints (in-sample); only 2008–2010 are test years. LSTM omitted (categorical embeddings not saved).